Appendices
Pollinator decline reveals uneven impacts of climate-driven biodiversity loss

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Appendix 1. Extended Methods, Materials and Data Processing

All data and codes are available at: https://github.com/gabrielemansi/Macroeconomic-impacts-of-pollination-loss-in-the-EU.git

Baseline pollination scenarios

To build the pollination potential baseline (2020-2070), the initial level of Pollination Potential for the year 2020 relies on data from the INCA (2018) potential (Vallecillo et al., 2022), normalized to a 0–1 scale.
The KIP-INCA framework (Vallecillo et al, 2018; MAES, 2020) developed a set of ecosystem services accounts (MAES, 2020) including crop pollination. The service is provided by the functional match between pollination potential, defined as the ecosystem's ecological suitability to support wild insect pollinators, and pollination demand, defined as the spatial extent of pollinator-dependent crops (the use area).

For this project we used estimates of pollination potential provided in "Ecosystem services accounts: maps for 7 ecosystem services produced in KIP INCA 2018 updates" (Vallecillo et al., 2022), available at (https://data.jrc.ec.europa.eu/dataset/4cbd7c1e-6512-4ebe-8ca5-e08209cc3efb , file at "\CROP_POLLINATION\potential\potential_categories.tif") .

The data is provided in a GeoTIFF images with the following characteristics:
Size: Image size: 4009, 4045 - Pixel Size: 1000, -1000
CRS: EPSG:3035 - ETRS89-extended / LAEA Europe - Projected
Extent: Corner Coordinates:

- Upper Left 2549000.0000000000000000,5447000.0000000000000000
- Lower Left 2549000.0000000000000000,1402000.0000000000000000
- Upper Right 6558000.0000000000000000,5447000.0000000000000000
- Lower Right 6558000.0000000000000000,1402000.0000000000000000

Unit : meters
The file includes the following bands: Year 2000, Year 2006, Year 2012, Year 2018

- min value: 1
- max value: 4

Description: Pollination potential is expressed with a categorical variable (1=None, 2=low, 3=medium, 4=high). The assessment of pollination potential is based on an indicator of the environmental suitability to support wild insect pollinators. It integrates two different models:
an Expert-Based Model for solitary bees and a Species Distribution Model for bumblebees.

Vallecillo et al., 2018 data classifies EU land surface into categories of pollination potential (1=None, 2=low, 3=medium, 4=high) based on a highly detailed assessment of land-cover composition and land-use patterns, aimed at identifying habitat elements suitable for pollinators. The assessment is based on an indicator of the environmental suitability to support wild insect pollinators which integrates two different models: an Expert-based Model (EBM) and a Species Distribution Model (SDM). More specifically, they build upon previous work undertaken by JRC staff, which has resulted in an EBM with a spatial resolution of 1 ha (100 x 100 m grid-cell) (Zulian et al., 2013), and a SDM based on bumblebee records, with a spatial resolution of 100 km2 (10 x 10 km) (Polce et al., 2013). Each of these approaches has some strengths and weaknesses: the EBM for instance, has the advantage of being able to account for the effect of detailed local information, such as the presence of wild flower edges between crop-fields, or other small patches of habitat suitable for pollinators. The EBM, however, might fail to reflect the environmental suitability for poorly known species, or to capture environmental characteristics that can modify the expected suitability (e.g. climatic differences) or, again, it might not be able to predict species richness. The SDM, on the other hand, has the advantage of being informed by actual species records, but it is constrained by the spatial and temporal resolution of these records. Hence, the SDM might fail to capture the effect of local landscape elements, if their accuracy is greater than what is available for the species records. By integrating the EBM and the SDM approach, therefore, they are able to better reflect the environmental suitability to support wild insect pollinators (and, hence, the pollination potential). Both models provide a 'suitability score' between 0 and 1 for each grid-cell. The 'suitability' is often interpreted as the 'capacity of the environment to support insect pollinators' (EBM) or the 'probability of occurrence of insect pollinator' (SDM). The EBM-suitability is based on experts' knowledge of the species ecology, solitary bees in our case. The SDM-suitability on the other hand, is derived through, e.g., statistics or machine learning techniques, which are used to characterise the 'quality' of the environment where species are recorded. In simple words, within a SDM, the relations between the environmental variables characterising the species' sightings, bumblebees in our case, are used to predict the environmental suitability across the area of interest. Since the SDM is informed by the sightings of species, it can capture complex relations among the different variables characterising the environment (e.g. land use and land cover, climate, etc.) where species are found. Environmental suitability is interpreted as a proxy for the 'probability of occurrence' for a species, therefore, by defining a threshold it is able to distinguish potential presence from absence (Liu et al., 2005). The JRC has produced individual SDMs for 47 bumblebee species, as well as a SDM resulting from aggregating all single SDMs. The score of the aggregated SDM is the mean of the single species' model for all species predicted to be present in a given cell, with ~ 76 ~ equal weights across species (weight =1/N, where N is the number of species present in a given cell). The original models by Zulian et al. (2013) and Polce et al. (2013) were adapted to meet the requirements of the accounting, such as the need to rely on datasets regularly updated. The outputs of the updated models were then averaged to estimate the potential availability of wild insect pollinators to relevant crop groups: the pollination potential.

Theme	Year		
	2000	2006	2012
LULC – Dynamic dataset	CORINE Accounting Layer 2000	CORINE Accounting Layer 2006	CORINE Accounting Layer 2012
Roads – Static dataset	Road network from TeleAtlas 2006 version (major roads only, corresponding to road category 0, 1 and 2); if a more recent version becomes available, it can be used from 2012 onward.		
Climate data – Dynamic dataset	Gridded Meteorological data from Agri4Cast[17] (Mean air temperature and total global radiation used for the Expert-Based model) and E-OBS[18] (minimum and maximum air temperature, sum of precipitation are used to compute the bio-climatic variables used for the Species-Distribution model - These variables were already computed and during the model calibration phase, and the most relevant one were selected.		
Species records – Static dataset	Bumblebee records from Atlas Hymenoptera[19]: • 47 species selected from ca. 60 (excluded species with very few records) • 10 x 10 km grid • 1991 to 2014		

Input data for the spatial indicator of pollination potential (indicator of potential supply of wild insect pollinators),from " Ecosystem services accounts: maps for 7 ecosystem services produced in KIP INCA 2018 updates".
Source: Vallecillo et al., 2018

The final model output is a dimensionless indicator of environmental suitability that is used to delineate service providing areas (SPA) with different suitability for pollinators: (1) High: environmental suitability above 0.3 ; (2) Medium: environmental suitability between 0.3 and 0.2; (3) Low: environmental suitability between 0.2 and 0.1; (4) None: environmental suitability below 0.1 .

To ensure consistency of our assessment with respect to other literature, we compared the Pollination Potential from our assessment (INCA 2018 normalized to 1) against the 2015 Pollination Sufficiency baseline from vonJeetse et. al 2023 (variable "pollSuff_2015_Ref_10s.tif").
Their study employs the coupled MAgPIE-SEALS modeling system to assess fine-scale changes in pollination sufficiency and soil loss from water erosion under four exploratory (what if) land-use scenarios projected for 2050.
Pollination Sufficiency Definition and Modeling Pollination sufficiency is defined as the amount of semi-natural habitat within typical foraging distances of wild pollinator communities surrounding cropland. It serves as a proxy for both wild pollination supply and configurational landscape

heterogeneity—an important driver of biodiversity and other regulating ecosystem services such as biological pest control. The modeling framework integrates spatial and ecological parameters to simulate interactions between pollinator habitats and crop dependencies over time. Calculations are performed at a 300 m × 300 m resolution using a landscape-based indicator adapted from Chaplin-Kramer et al. (2015), based on two components:

- Habitat Definition: Includes forest, non-forest vegetation, and grassland—assumed to support wild pollinators.
- Foraging Radius: A 2 km buffer is applied around each cropland pixel. If ≥30% of this area is semi-natural habitat, the pixel receives a sufficiency score of 1. If below 30%, the score is scaled linearly between 0 and 1.

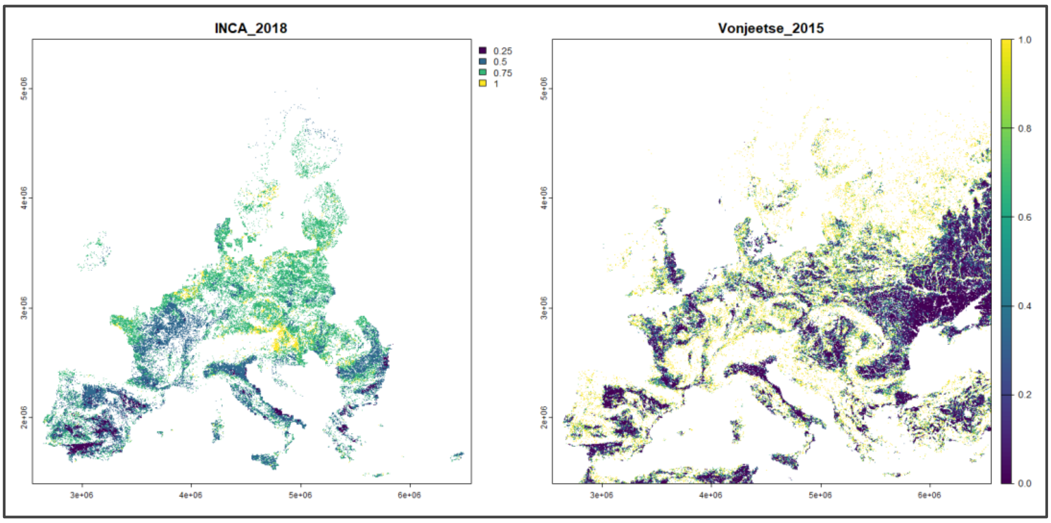
These scores quantify pollination supply and landscape heterogeneity, with each cropland pixel assigned a value from 0 to 1:

- 1 = sufficient habitat (≥30%)
- (0 – 1) = proportional habitat (<30%)

Scenario Design and Assumptions The reference scenario—Business-as-Usual (BAU)—follows the SSP2 "middle-of-the-road" pathway, assuming moderate population and income growth alongside current national land conservation policies. Climate data are used for baseline biophysical simulations but do not factor into future pollinator dynamics; climate change impacts such as altered rainfall or temperature are excluded from the pollination modeling.

In the baseline data (base year 2018 and 2015), the two metrics proved to be broadly comparable, with closely aligned mean values across the EU.

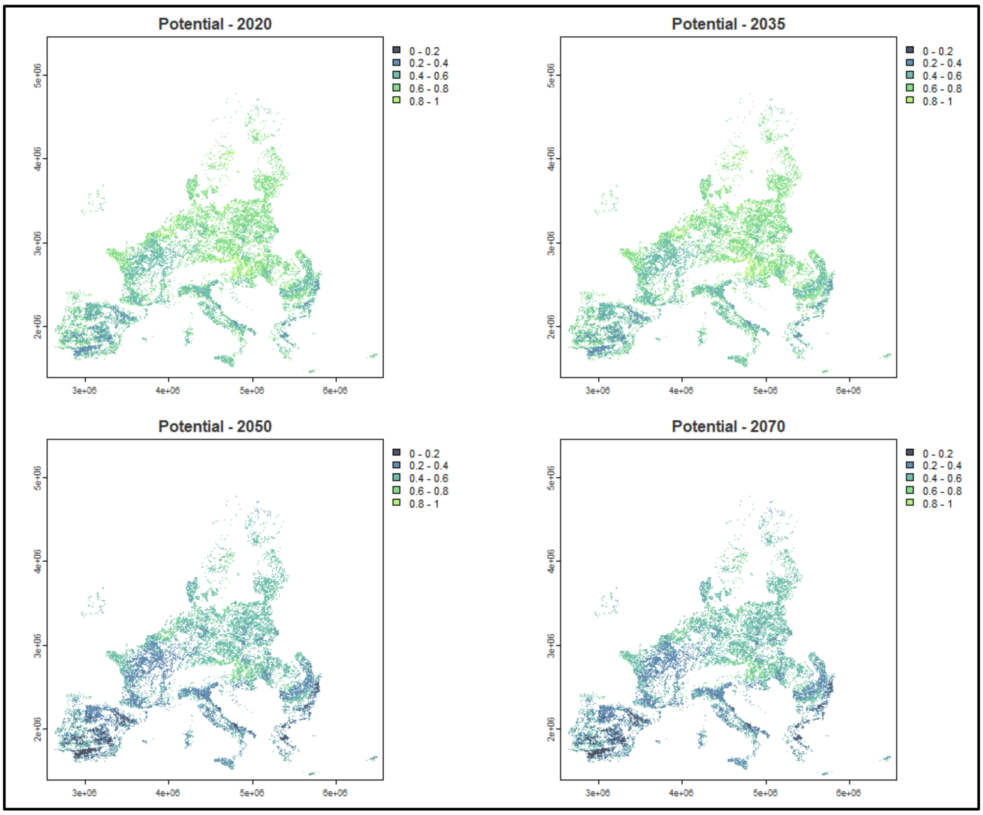
	Layer	Mean	Dataset
1	2018	0.64871	INCA_2018
2	2015	0.73535	vonJeetse



Comparison between vonJeetse et. al 2023 pollSuff_2015_Ref_10s and INCA_2018 pollination potential normalized to 0-1.

The INCA pollination potential represents a snapshot of current ecological conditions and does not explicitly incorporate temporal trends. Likewise, species distribution models (SDMs) generally omit several relevant stressors, including pollutant exposure (see Appendix 1). Introducing an explicit

temporal degradation complements the static baseline and better captures observed declines not otherwise represented in the modelling framework. In this context, the 1991–2020 trend is interpreted as broadly independent of climate-driven effects and is used to construct the no-climate-change baseline.
Future changes are modelled by scaling down the 2020 baseline potential up to 2050 in line with observed historical trends in pollination loss. A -1\% annual decline in pollination potential is imposed up to 2035, under the assumption that the observed negative trajectory persists throughout the present decade and reaches its maximum intensity around that year. From 2035 onwards, the rate of decline is assumed to decelerate progressively, following a linear tapering path until it converges to a 0\% change rate by 2050. As a consequence, in the absence of climate change effects, no further decline is projected between 2050 and 2070, implying a stabilisation of pollination potential in the long run under the baseline scenario.
The broader body of literature reviewed (see: Appendix 2), together with expert consultation with the INCA team at the Joint Research Centre (JRC), supports the adoption of a -1% annual decline as a robust and defensible assumption for the EU. This parameter is consistent with estimates developed within the EU Science and Technology for Pollinating Insects (STING) project (Potts et al., 2024), ensuring methodological coherence with the most recent EU-level scientific assessments and reinforcing its empirical plausibility within the modelling framework. This is also consistent with the expected long-term trend of agro-ecosystem loss reported by the EU (2023), and the 1991-2020 trend of the EU Grassland Butterfly Index (-1.23% per year), a widely used proxy for pollinator-related biodiversity change due to its long temporal and spatial coverage and sensitivity to the same environmental pressures that affect wild pollinators (Eurostat, 2025; Van Swaay, 2025). These include land-use change, habitat loss and fragmentation (resulting, among other factors, from agricultural intensification and monocultures, urbanization and infrastructure development, removal of hedgerows and field margins, loss of semi-natural grasslands, and decline of wild flowering plants), as well as exposure to environ-mental pollutants such as pesticides (especially systemic insecticides) (Nicholson et al., 2024) and fungicides, and the presence of invasive species, all factors that are more prevalent in intensive agricultural systems. Since INCA pollination potential and SDMs in general do not account for many of these sources of impact (such as pollutants), imposing historical trend over baseline future pollination allows us to capture real-world pollination decrease trends not specifically modelled both in the initial data and in the SDM projections used for climate impacts.
In our modelling framework, therefore, a -1\% annual decline in pollination potential is imposed up to 2035, under the assumption that the observed negative trajectory persists throughout the present decade and reaches its maximum intensity around that year. From 2035 onwards, the rate of decline is assumed to decelerate progressively, following a linear tapering path until it converges to a 0% change rate by 2050.
The assumption is deliberately conservative, intended to avoid overstating long-term losses under substantial uncertainty in ecological trajectories and policy effectiveness. In particular, it reflects the fact that existing policy interventions, such as the EU Pollinators Initiative and earlier reforms under the Common Agricultural Policy (CAP), may not yet have fully translated into observable trends. Extrapolating current declines over extended horizons would otherwise rely on a strong and potentially unjustified assumption of persistence, with a risk of systematic overestimation of long-run impacts. As we expect that the historical trend does not capture significant climate change impacts, the pollination potential assessed in this way is used as a baseline for the no-climate change scenario (scenario that holds future climate constant).



Projected pollination potential (0-1) under the baseline scenario (2020-2070) with no climate change.

Climate change scenarios

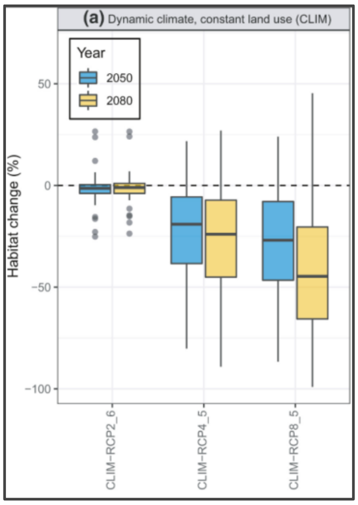
Projected declines in pollination potential due to changing climate are derived from an ensemble of state-of-the-art multi-model, multi-scenario species distribution models (SDMs) for key pollinator taxa (wild bees, hoverflies, butterflies, beetles), based on the most recent and robust data currently available (a detailed description of data sources is provided in Appendix 1). This framework enables the assessment of multiple climate scenarios (RCP2.6, 4.5, 6.0, 7.0, 8.5) and socioeconomic pathways (SSP1, 2, 3, 5) to project climate-driven impacts on pollinators up to 2070 (Figure). Data sources were selected to ensure full consistency with the KIP-INCA and Status and Trends of European Pollinators (STEP) frameworks. By mapping baseline pollination potential together with spatially explicit projections of climate change impacts on pollinators' habitat suitability, we estimate the expected future pollination potential and its percentage change relative to the baseline (2020–2070) under different climate change (RCPs) and land use (SSPs) scenarios , at a very high spatial resolution of 1 km × 1 km.

Scenario name	Land Use	RCP	Tot
PREST-CLIM	Constant	2.6 - 4.5 - 8.5	3
IIASA-CLIM	Constant	2.6 - 4.5 - 7.0 - 8.5	4
PREST-1	SSP1	2.6 - 4.5	2
PREST-3	SSP3	4.5 - 8.5	2
PREST-5	SSP5	8.5	1
IIASA-2	SSP2	2.6 - 4.5 - 7.0	3
GHSB-1	SSP1	2.6	1
GHSB-3	SSP3	6.0	1
GHSB-5	SSP5	8.5	1
All land-use / RCP combination considered			18

The pollination scenarios considered in this study as a combination of land use variables and climate radiative forcing (RCPs) used by each species distribution model (SDM)

To model **PREST-CLIM** and **PREST-1,3,5** scenarios, data from Prestele et al., 2021 are used to proxy the climate change impacts on pollinators affecting the baseline potential according to three climate change scenarios, that is the representative concentration pathways (RCPs) corresponding to distinct shared socioeconomic pathways (SSPs): SSP 1-2.6, SSP 1-4.5, SSP 3-4.5, SSP 3-8.5 and SSP 5-8.5 (+2.6, +4.5, +8.5 W m−2 of radiative forcing by 2100 relative to pre-industrial values, corresponding to low, medium and high greenhouse gas emissions, respectively).
The choice was made because of data consistency with the INCA and MAES framework and Status and Trends of European Pollinators EU STEP project. Prestele et al., 2021 project the potential distribution of 47 European bumblebee species in 2050 and 2080 from existing European-scale distribution maps, based on a set of climate and land-use futures simulated through a regional integrated assessment model and consistent with the RCP–SSP scenario framework. Baseline bumblebee distributions were based on Species Distribution Model (Polce et al., 2018) and adjusted to be consistent with land-use data from the projections. Future climate and land-use data were taken from CMIP5 climate model simulations (Taylor et al., 2012) and the IAP2 (Harrison et al., 2019), respectively. They applied the SDMs to the time periods representative for the years 2050 and 2080 to obtain estimates of the future distributions of the 47 bumblebee species.
The potential distribution of 47 bumblebee species and SDM model coefficients were made available by Polce et al. (2018). The distribution maps indicate for each species (1) the probability of occurrence and (2) presence/absence at a spatial resolution of 10 x 10 km, based on a maximum entropy (MaxEnt) modeling approach (Phillips et al., 2006) that establishes a functional relationship between species occurrence and 22 environmental predictor variables.
Various levels of climate change were considered through the scenario framework of the IPCC, combining representative concentration pathways (RCPs) with shared socioeconomic pathways (SSPs) (O'Neill et al., 2017; van Vuuren et al., 2011). Their findings show that direct climate impacts, although variable across species, dominate responses for most species, especially under high-end climate change scenarios (up to 99% range loss).
For our constant land use PREST-CLIM scenarios, we use data from the dynamic climate and constant land use (CLIM) experiment, where results are not determined by SSPs' related demand and land use

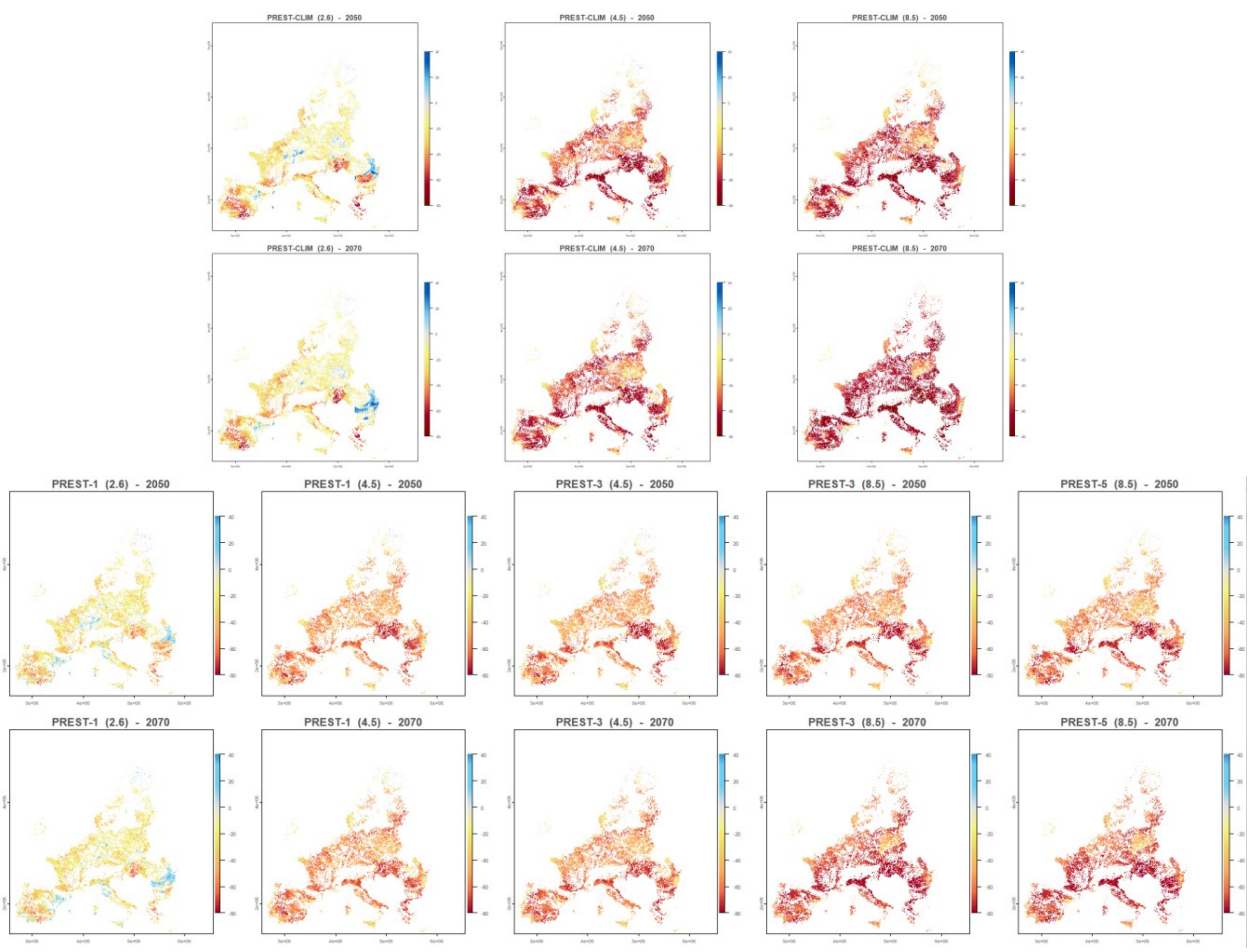
change (Figure below), but only by different levels of radiative forcing (RCPs), so that in our modelling framework, future land use is held constant from FAO CROPGRIDS (2020) datasets.



EU averaged habitat change for pollinators species under climate change and constant land use in 2050 and 2080. Source: Prestele et al., 2021

For the PREST-1,3,5 scenarios, we use data from dynamic climate and dynamic land use (COMB) experiment. Given a certain level of climate change (RCP2.6, RCP4.5, RCP8.5), including dynamic land-use change (COMB modeling experiment) has varying impacts on the species' response across land-use scenarios. In a world consistent with RCP2.6 climate in 2080, land use following the SSP1 storyline results in an average additional habitat change of −7.50%, which is largely driven by large losses of individual species. In contrast, SSP4 land-use changes almost offset average habitat losses from climate change (+2.09%). In the COMB-RCP2.6-SSP1 scenario 22 species lose additional habitat compared to the respective CLIM-RCP2.6 scenario, while 25 species gain habitat. In COMB-RCP2.6-SSP4 19 species lose habitat, while 28 gain habitat; but, more importantly, it is different species that are affected in the one or other direction. SSP1 land use leads to strong additional habitat losses for rare species (−31.21% for the lower quartile) while common species tend to benefit (+2.35% and +1.15% for the third and upper quartiles). SSP4 land use, in contrast, attenuates habitat losses for rare species (+11.37% for the lower quartile) and common species experience additional losses (−1.66% for third and −0.83% for the upper quartile). For 35 out of 47 species, the choice of land-use scenario has an opposite effect on the net change to suitable habitat area (i.e., habitat loss in one land-use scenario contrasts with habitat gain in the other) and for 24 species it has the potential to change the sign of the combined signal compared to the CLIM modeling experiment. Importantly, differences between land-use scenarios do also exist in RCP4.5 and RCP8.5 worlds in 2080, although the importance of land-use decreases with higher levels of climate change.

Overall, the data are spatially referenced and capture both increases and decreases in pollinator distributions as determined by projected regional climate conditions under the various SSP–RCP combinations. Consequently, some regions are expected to experience gains in pollination potential, while others will face substantial declines. This reflects the inherently local nature of climate impacts: evidence shows that many wild pollinator species are likely to shift towards northern ranges, whereas most bumblebee species are projected to undergo significant range contractions (Potts et al., 2015). By combining baseline pollination potential with spatially explicit projections of climate-induced changes in pollinator distributions, we estimate future pollination potential and its percentage change relative to the baseline (2020–2080) across multiple climate scenarios (RCP 2.6, 4.5, and 8.5), at a high spatial resolution of 1 km × 1 km.

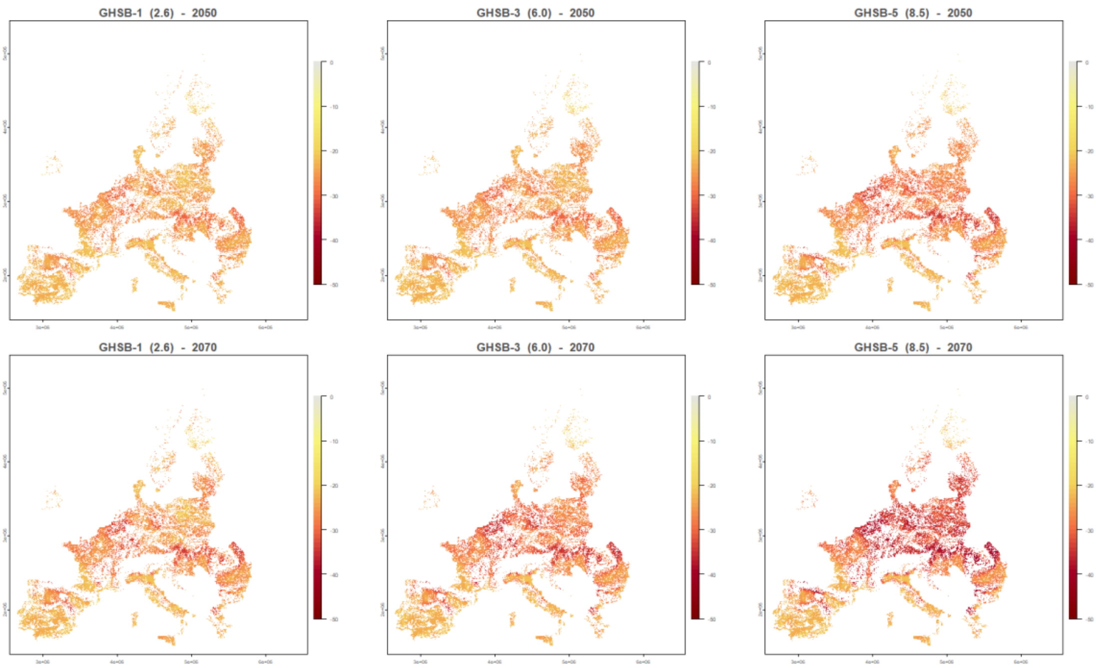


Each panel shows the average projected pollination potential percentage change in 2050 and 2070 with respect to 2020 under PREST-CLIM and PREST-1,3,5 scenarios.

To model **GHSB-1,2,3** climate change scenarios, data from Ghisbain et al., 2024 are used to proxy the climate change impacts on pollinators affecting the baseline potential, according to three climate change scenarios, that is the representative concentration pathways (RCPs) corresponding to three distinct shared socioeconomic pathways (SSPs): SSP 1-2.6, SSP 3-6.0 and SSP 5-8.5 (+2.6, +6.0 and +8.5 W m^{-2} of radiative forcing by 2100 relative to pre-industrial values, corresponding to low, medium-high and high greenhouse gas emissions, respectively).

Authors analyse the past, present and future ecological suitability of the European continent for 46 bumblebee species on the basis of a set of harmonized climate, land use and human population data obtained through the Inter-Sectoral Impact Model Intercomparison Project Phase 2b (ISIMIP2b) for the last century, the present period and for upcoming years to 2080. The observational dataset comprises 401,046 unique georeferenced occurrence records of bumblebee presence split into 'past' (1901–1970 inclusively) and 'present-day' time periods (2000–2014 inclusively; Methods). To model the species–environment relationships between bumblebee occurrence records and the set of ISIMIP2b variables, they train ecological niche models on georeferenced data corresponding to each period using a boosted regression tree (BRT) approach. The resulting predictive probability is here interpreted as a measure of ecological suitability given local environmental conditions and varies between 0 (unsuitable environmental conditions) and 1 (highly suitable environmental conditions). In addition to the computation of commonly used area under the receiver operating characteristic curve (AUC) metric, they assessed the predictive performance of the ecological niche models by computing a prevalence-pseudoabsence-calibrated Sørensen's index. Using the same index, they then evaluated the predictive performance of our models by assessing the ability of ecological niche models trained on past occurrence and environmental data to predict present-day distributions when projected on present-day environmental data, as well as the ability of ecological niche models trained on present-day occurrence and environmental data to predict past distributions when projected on past environmental data.

In addition to training and mapping BRT models with the past and present-day occurrence data and environmental conditions, models trained on present-day data are further used to project the changes in ecological suitability maps in the next decades. For this purpose, they use environmental layers projected for three time windows (2021–2040, 2041–2060 and 2061–2080) according to three climate change scenarios, that is the representative concentration pathways (RCPs) corresponding to three distinct shared socioeconomic pathways (SSPs): SSP 1-2.6, SSP 3-6.0 and SSP 5-8.5 (+2.6, +6.0 and +8.5 W m−2 of radiative forcing by 2100 relative to pre-industrial values, corresponding to low, medium-high and high greenhouse gas emissions, respectively). They used three different global change trajectories instead of a single one to account for uncertainty surrounding future global changes, which are contingent on societal decisions.



Each panel shows the average projected pollination potential percentage change in 2050 and 2070 with respect to 2020 under GHSB-1,2,3 scenarios.

To model **IIASA-CLIM** and **IIASA-2** climate change scenarios, we rely on IIASA analysis on habitat suitability under climate and land-use scenarios from ACCREU project deliverable D4.2.[1] (Gerber G., Palazzo A. et al., 2025). They utilized individual species distribution models (SDMs) for key pollinator groups Lepidoptera (Butterflies) and Hymenoptera (Bees), representing, along with Coleoptera and Diptera (not modelled due to data limitations), the dominant taxa providing pollination services to crops

[1] ACCREU (EU), 2025: Deliverable D2.4 (Impacts on Ecosystems & Biodiversity). GA number: 101081358; Funding type: HORIZON-CL5-2022-D1-07-two-stage.

in Europe (Moldoveanu et al., 2024; Potts et al., 2016; Reverté et al., 2023; Wardhaugh, 2015). The analysis focused on species that had (1) sufficient species occurrence records (Chapman et al. 2025), and (2) species-specific available habitat preference information (IUCN Red List database or EEA databases). They excluded non-native species (based on Nature Directives listings) and fully aquatic species (to focus on terrestrial and semi-aquatic taxa), focusing on species that are listed in the EU Habitat and Birds directive, as well as species of relevance to the EU pollinator initiative. For each species, they extracted and formatted habitat preferences and converted them to spatial priors aligned with the land-use classification system. For additional nuance, they also incorporated weights into habitat preferences based on their suitability classification: "Suitable"/"Preferred" habitats received full weight (1.0), "Marginal"/"Occasional" habitats received reduced weight (0.5). Finally, they incorporated species-specific suitable elevation ranges where available (Brooks et al., 2019). Otherwise, the full elevation range was used.

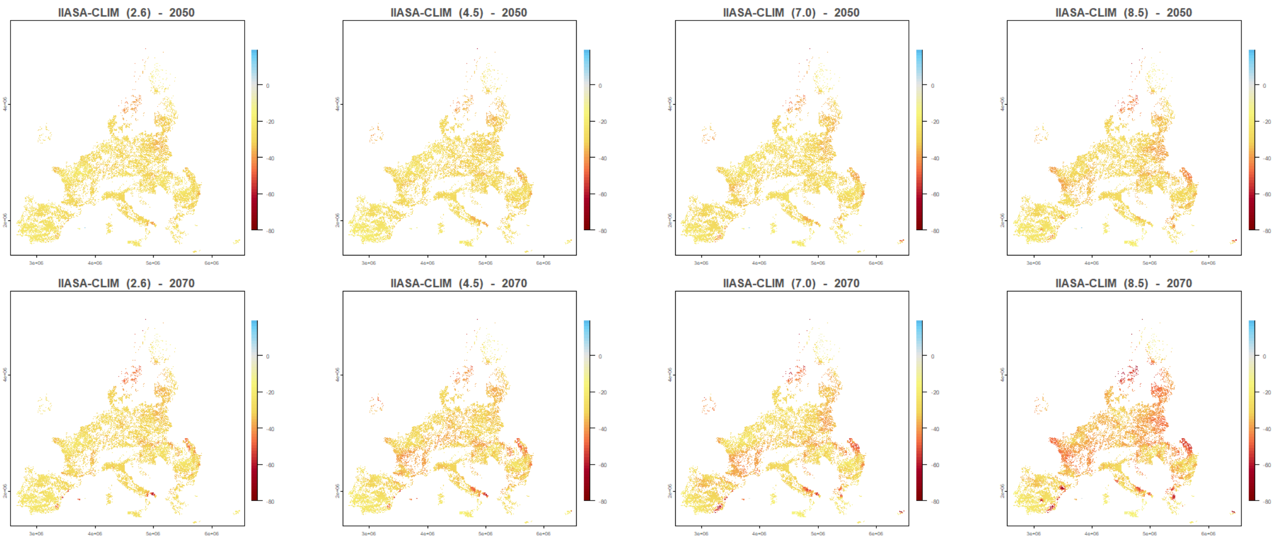
Pollinator Group	Species count	% of Total Pollinators	Represented Genus (Species count per genus)
Hymenoptera (Bees)	96	62.75	*Andrena (10); Bombus (11); Chelostoma (1); Colletes (2); Dufourea (1); Epeolus (2); Halictus (2); Hoplitis (3); Hylaeus (2); Lasioglossum (2); Macropis (1); Megachile (3); Nomads (12); Osmia (1); Panurgus (2); Sphecodes (1); Stelis (1)*
Lepidoptera (Butterflies)	57	37.25	*Anthocharis (1); Apatura (1); Aricia (1); Arytrura (1); Cacyreus (1); Clossiana (1); Coenonympha (6); Colias (1); Cupido (1); Danaus (1); Erannis (1); Erebia (20); Eriogaster (1); Euchloe (2); Euphydryas (2); Gegenes (1); Glyphipterix (1); Hipparchia (7); Laeosopis (1); Lampides (1); Lasiommata (1); Leptidea (1); Leptotes (1); Lignyoptera (1); Lopinga (1); Lycaena (3); Lysandra (1); Melanargia (2); Melitaea (4); Nymphalis (2); Oeneis (1); Papilio (2); Parnassius (2); Pelopidas (1); Pieris (1); Polyommatus (4); Pontia (1); Proserpinus (1); Pseudophilotes (2); Pyrgus (6); Vanessa (3); Xestia (1); Xylomoia (1); Zerynthia (1)*
Total	**153**	**100 %**	

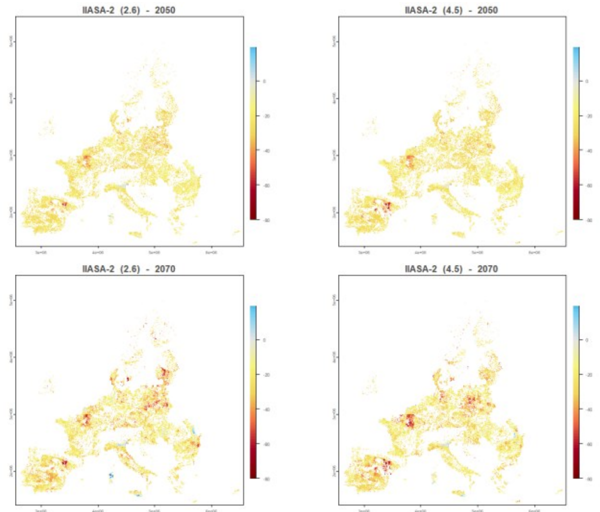
Summary of wild insect pollinator groups (Hymenoptera, Lepidoptera) considered in the IIASA-CLIM and IIASA-2 scenarios. Source: ACCREU (EU) Deliverable D2.4 (Gerber G., Palazzo A. et al., 2025)

The SDMs provide species-specific habitat availability indices, represented as a fraction of suitable habitat per 10 km 2 grid cell across Europe for different climate and adaptation scenarios from 2015 to 2095 in 10-year intervals. From these, they calculated species-weighted habitat availability across all pollinator species for each 10 km 2 grid cell, providing a measure of overall pollinator potential (0-1). The analysis captured the combined impact of future climate and land use on individual species distributions by coupling a species distribution modelling approach with the ibis.iSDM modelling tool (Jung, 2023a) and refinement of suitable habitat (Brooks et al., 2019; Pereira et al., 2024; Visconti et al., 2024, 2016). By refining a species distribution by its species-specific habitat requirements, they capture the availability of natural and modified land directly available to the species. This enables them to capture both climatically driven shifts in the distribution of European species and their attributed LULUCF impacts in terms of land-use change. This integrated approach allows for a better understanding of how land-use changes and management practices affect species habitat suitability beyond what climate-only SDMs can provide.

The IIASA-CLIM scenarios are based on on the ‘climate_only_mean_suitability’ experiment which relies on the 2015/2025 baseline land use data and future modelled GCMs projections from ISIMIP3b climate forcing data (DOI: https://zenodo.org/records/13259644). In the data used for the IIASA-2 scenarios, land use projections from the downscaled GLOBIOM land use maps of the ACCREU scenarios (Deliverable 2.2) are also used.

They estimated the corresponding (matched GCM–RCP) future climate-only species projections using climate forcing data from ISIMIP, which were temporally aggregated to a decadal scale. Specifically, they trained and projected for each species and GCM-RCP combination, a total of 4 different Bayesian (e.g., regularized point process regressions) and machine learning (e.g., Extreme Gradient Descent Boosting). Furthermore, they constructed for each model 3 different folds utilizing spatial block cross-validation approaches (Roberts et al., 2017). Contemporary and future predictions were first averaged through a weighted ensemble calculation, by maximizing a True Skill Statistic (TSS) calculated on a threshold map, first across folds and then across GCMs. To constrain future projections, they created a 'mcp_dispersal' variant, which constrains the projected habitat suitability by the minimum convex polygon covering all observations and a negative exponential kernel with an average dispersal estimate per taxonomic group following Thuiller et al. (2019). All modelling was conducted using the ibis.iSDM R-package (Jung, 2023a), with hyperparameters and model contributions varied depending on data availability per species to further limit overfitting. The resulting projections capture the climatic niche (unit 0-1) of a species for each SSP and adaptation. Finally, for each species, they calculated within the current and future climatic niche the amount of suitable habitat, integrating information on downscaled land-use change for each SSP-RCP scenario combination and species-specific habitat preferences. This was done using the ibis.insights R package (Jung, 2023b), adopting an approach by Visconti et al. (2016). They note that the linkage between species ranges and land use is dependent on the accuracy and availability of habitat preference information. Particularly for permanent croplands, full impacts on habitat availability can be underestimated, as species-specific coefficients are not always available.



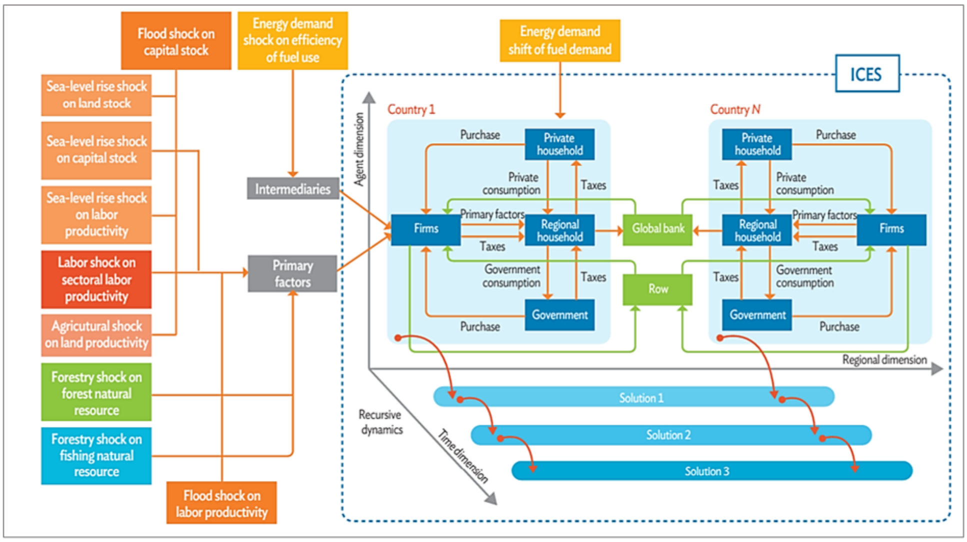


Each panel shows the average projected pollination potential percentage change in 2050 and 2070 with respect to 2020 under IIASA-CLIM and IIASA-2 scenarios.

Aggregation of pollination shocks to GTAP sectors

The projected gridded percentage changes in yield for each crop and climate scenario are mapped and aggregated at NUTS levels (0 to 3) and then used as land-productivity shocks in the macroeconomic model. Macroeconomic impacts are then estimated using a regionalized version of the ICES Computable General Equilibrium (CGE) model (Standardi et al. 2023; van der Wijst et al. 2023) based on GTAP11 database and parameters (Aguiar et al., 2022), which provides detail for 122 regions down to NUTS-2 level across the EU27, while the rest of the world is represented as a macro-regional aggregate.

The theoretical structure of the CGE model is based on the GTAP model (Hertel, 1997). Representative optimizing agents (households, firms) maximise an objective function subject to a specific constraint. Households maximise the utility flow from consumption subject to a budget constraint. Firms maximise profits subject to a technological constraint. Labour and capital are fully employed and mobile across sectors but not between regions. Investment is saving driven, and trade between subnational regions follows the Armington assumption (imperfect substitutability between domestic and foreign product). The model is recursive-dynamic, this implies that agents have myopic expectations (no perfect foresight) and the investment computed in the previous time period is added to the depreciated capital in the subsequent time step.



ICES model and channels of implementation of climate change-related shocks. Source: Campagnolo et al., 2026

To perform the macroeconomic assessment of our exercise, the GTAP11 database agricultural sub-sectors entering the CGE model have been aggregated in crop categories as follows:

	GTAP 11	GTAP 11 description	Sect aggr	Paper description
1	pdr	Paddy rice	gro	Cereals, wheat, rice
2	wht	Wheat	gro	Cereals, wheat, rice
3	gro	Cereal grains nec	gro	Cereals, wheat, rice
4	v_f	Vegetables, fruit, nuts	v_f	Vegetables, fruit, nuts
5	osd	Oil seeds	osd	Oil seeds
6	c_b	Sugar cane, sugar beet	c_b	Sugar cane, sugar beet
7	pfb	Plant-based fibers	ocr	Other crops, fibers
8	ocr	Crops nec	ocr	Other crops, fibers

Since the resulting agricultural sector in the CGE is relatively aggregated, we averaged the projected yield percentage changes based on the relative weight of each crop cultivated in each region, aligning them with the crop categories used in the model. This approach is also convenient because the GTAP crop categories generally correspond to similar levels of pollination dependence; for example, Vegetables and fruits have, on average, high dependence, Oil seeds medium, while Cereals, Wheat and Rice low to negligible dependence. To do this, we compute a weighted average in which the weight of the specific crop in each grid cell within a NUTS region is proportional to the number of hectares cultivated in that cell. These results are then further aggregated to match the crop categories used in the economic model, taking into account the relative contribution of each crop within the NUTS region of interest.

With the exception of the CLIM scenarios, which are based on a constant 2020 land use exercise, we incorporate projected crop and crop category-specific land-use changes from Goebel and Britz (2025), to account for future land-use changes and crop expansion under different SSPs (Riahi et al., 2017, van Vuuren et al., 2014) at the highest resolution available. Projections are made so that they ensure full consistency with our modelling framework, as they rely on a Computable General Equilibrium platform built on the GTAP v11 Power Database (Aguiar et al., 2022; Chepeliev, 2023; GTAP, 2023). To do so, we exogenously imposed the land use trajectories, both in the baseline rounds and in the impact scenarios, by working on the QOEs variable in the model.

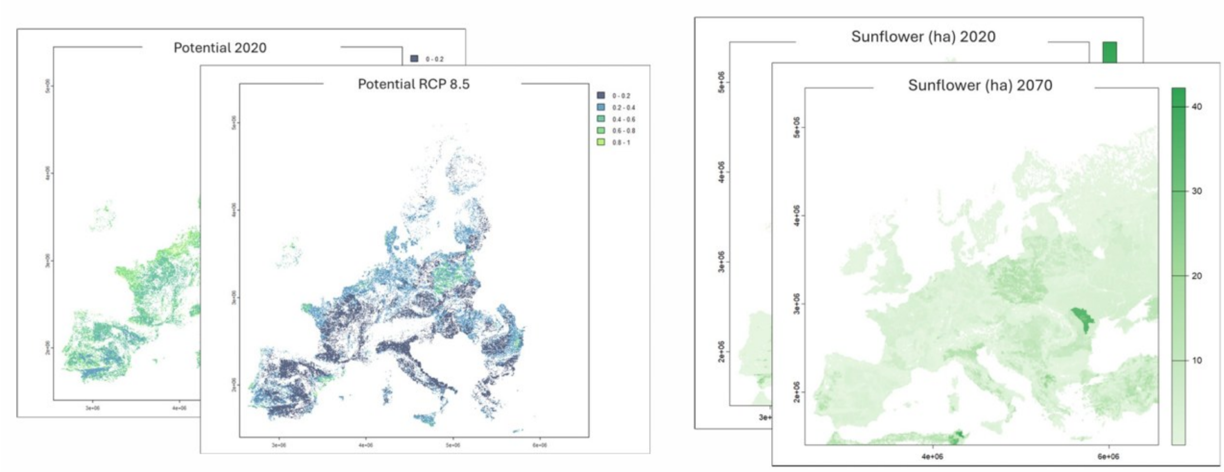
The table below shows the aggregation matches performed for the economic impact projections between:

(i) the crops and crop categories (Crop) from FAO CROPGRIDS (2020) with a dependency rate to wild pollination (Dep. Rate) above 0.05;
(ii) the corresponding sector (GTAP11 sec) in the CGE model and;
(iii) the land use categories (Land-Use) from the SSPs projections from Goebel and Britz (2025).

	Crop	GTAP 11 sec		Dep. Rate	Land Use
1	alfalfa	ocr	Crops nec	0.95	ocro
2	almond	v_f	Vegetables, fruit, nuts	0.65	ofru
3	aniseetc	ocr	Crops nec	0.05	ocro
4	apple	v_f	Vegetables, fruit, nuts	0.65	appl
5	apricot	v_f	Vegetables, fruit, nuts	0.65	ofru
6	artichoke	v_f	Vegetables, fruit, nuts	0.25	oveg
7	asparagus	v_f	Vegetables, fruit, nuts	0.05	oveg
8	avocado	v_f	Vegetables, fruit, nuts	0.65	oveg
9	banana	v_f	Vegetables, fruit, nuts	0.05	banp
10	bean	ocr	Crops nec	0.05	leg
11	beetfor	ocr	Crops nec	0.05	ocro
12	berrynes	v_f	Vegetables, fruit, nuts	0.50	ofru
13	blueberry	v_f	Vegetables, fruit, nuts	0.65	ofru
14	broadbean	v_f	Vegetables, fruit, nuts	0.25	ofru
15	buckwheat	ocr	Crops nec	0.65	ocro
16	cabbage	v_f	Vegetables, fruit, nuts	0.05	oveg
17	cabbagefor	v_f	Vegetables, fruit, nuts	0.05	oveg
18	carob	v_f	Vegetables, fruit, nuts	0.25	oveg
19	castor	osd	Oil seeds	0.05	oso
20	cauliflower	v_f	Vegetables, fruit, nuts	0.05	oveg
21	cherry	v_f	Vegetables, fruit, nuts	0.65	ofru
22	chestnut	v_f	Vegetables, fruit, nuts	0.25	ofru
23	chickpea	ocr	Crops nec	0.25	leg
24	chicory	v_f	Vegetables, fruit, nuts	0.25	oveg
25	chilleetc	ocr	Crops nec	0.25	leg
26	citrusnes	v_f	Vegetables, fruit, nuts	0.65	citr
27	cotton	ocr	Crops nec	0.25	pfb
28	cowpea	ocr	Crops nec	0.05	leg
29	cranberry	v_f	Vegetables, fruit, nuts	0.65	ofru
30	cucumberetc	v_f	Vegetables, fruit, nuts	0.65	oveg
31	currant	v_f	Vegetables, fruit, nuts	0.05	ofru
32	date	v_f	Vegetables, fruit, nuts	0.05	ofru
33	eggplant	v_f	Vegetables, fruit, nuts	0.25	oveg
34	fig	v_f	Vegetables, fruit, nuts	0.65	ofru
35	flax	v_f	Vegetables, fruit, nuts	0.25	oveg
36	fornes	v_f	Vegetables, fruit, nuts	0.05	oveg
37	fruitnes	v_f	Vegetables, fruit, nuts	0.65	ofru
38	gooseberry	v_f	Vegetables, fruit, nuts	0.25	ofru
39	grape	v_f	Vegetables, fruit, nuts	0.05	grap
40	grapefruitetc	v_f	Vegetables, fruit, nuts	0.05	grap
41	greenbean	v_f	Vegetables, fruit, nuts	0.05	oveg
42	greenbean	ocr	Crops nec	0.25	leg
43	greencorn	gro	Cereal grains nec	0.05	maiz
44	greenonion	v_f	Vegetables, fruit, nuts	0.05	oveg
45	groundnut	osd	Oil seeds	0.05	oso
46	kiwi	v_f	Vegetables, fruit, nuts	0.95	ofru
47	legumenes	ocr	Crops nec	0.25	leg
48	lemonlime	v_f	Vegetables, fruit, nuts	0.05	citr
49	lettuce	v_f	Vegetables, fruit, nuts	0.05	oveg
50	linseed	osd	Oil seeds	0.25	oso
51	maize	gro	Cereal grains nec	0.05	maiz
52	maizefor	gro	Cereal grains nec	0.05	maiz
53	melonetc	v_f	Vegetables, fruit, nuts	0.65	ofru
54	melonseed	v_f	Vegetables, fruit, nuts	0.65	ofru
55	mixedgrain	gro	Cereal grains nec	0.05	ocer
56	mixedgrass	ocr	Crops nec	0.05	pfb
57	mustard	ocr	Crops nec	0.25	leg
58	nutnes	v_f	Vegetables, fruit, nuts	0.65	ofru
59	oilseedfor	osd	Oil seeds	0.25	oso
60	oilseednes	osd	Oil seeds	0.25	oso
61	okra	v_f	Vegetables, fruit, nuts	0.25	oveg
62	olive	osd	Oil seeds	0.05	olv
63	onion	v_f	Vegetables, fruit, nuts	0.05	oveg
64	orange	v_f	Vegetables, fruit, nuts	0.25	citr
65	peachetc	v_f	Vegetables, fruit, nuts	0.65	ofru
66	pear	v_f	Vegetables, fruit, nuts	0.65	ofru
67	peppermint	ocr	Crops nec	0.25	ocro
68	persimmon	v_f	Vegetables, fruit, nuts	0.05	ofru
69	pimento	v_f	Vegetables, fruit, nuts	0.25	ofru
70	pineapple	v_f	Vegetables, fruit, nuts	0.25	ofru
71	plum	v_f	Vegetables, fruit, nuts	0.65	ofru
72	poppy	ocr	Crops nec	0.25	leg
73	potato	v_f	Vegetables, fruit, nuts	0.05	pota
74	pulsenes	ocr	Crops nec	0.25	leg
75	pumpkinetc	v_f	Vegetables, fruit, nuts	0.95	oveg
76	quince	v_f	Vegetables, fruit, nuts	0.25	oveg
77	rapeseed	osd	Oil seeds	0.25	rape
78	rasberry	v_f	Vegetables, fruit, nuts	0.65	ofru
79	safflower	osd	Oil seeds	0.05	oso
80	sesame	osd	Oil seeds	0.25	oso
81	sisal	osd	Oil seeds	0.25	oso
82	sourcherry	v_f	Vegetables, fruit, nuts	0.65	ofru
83	soybean	ocr	Crops nec	0.25	soy
84	spinach	v_f	Vegetables, fruit, nuts	0.05	oveg
85	stonefruitnes	v_f	Vegetables, fruit, nuts	0.65	ofru
86	strawberry	v_f	Vegetables, fruit, nuts	0.65	ofru
87	stringbean	ocr	Crops nec	0.25	leg
88	sugarbeet	c_b	Sugar cane, sugar beet	0.05	c_b
89	sugarcane	c_b	Sugar cane, sugar beet	0.05	c_b
90	sunflower	osd	Oil seeds	0.45	oso
91	swedefor	v_f	Vegetables, fruit, nuts	0.05	pota
92	sweetpotato	v_f	Vegetables, fruit, nuts	0.05	pota
93	tangetc	v_f	Vegetables, fruit, nuts	0.05	oveg
94	taro	ocr	Crops nec	0.25	ocro
95	tea	ocr	Crops nec	0.25	ocro
96	tobacco	ocr	Crops nec	0.25	ocro
97	tomato	v_f	Vegetables, fruit, nuts	0.25	toma
98	tropicalnes	v_f	Vegetables, fruit, nuts	0.25	ofru
99	turnipfor	ocr	Crops nec	0.65	ocro
100	vegetablenes	v_f	Vegetables, fruit, nuts	0.65	oveg
101	vegfor	v_f	Vegetables, fruit, nuts	0.25	oveg
102	watermelon	v_f	Vegetables, fruit, nuts	0.95	ofru
103	yam	ocr	Crops nec	0.25	ocro

The dependency rates reported here measure the average impact on production of pollination loss, i.e. the fractional reduction in crop commercial yield associated with a loss of animal pollination (Klein et al., 2007; Bugin et al., 2022; Siopa et al., 2024). For example, a pollination dependency of 0.05, means that a 100% decrease in pollination results in a 5% decrease in commercial yield. The impact on production for a specific crop in a given area is therefore determined by the projected change in pollination service in the area where the crop is cultivated and by that crop's dependency on animal pollination. Dependency rates were taken from literature using multiple sources, partially considering also the indirect effect on crop quality when this significantly influence marketable yield (Klein, 2007;

Aizen et al., 2009; Bauer, 2016; Perrot et al., 2019; Bugin et al., 2022; Munir et al., 2020; Layek et al., 2023).



A representation of the spatial overlapping between pollination potential area (left) and demand area (right).

Appendix 2. Pollination current status and projected trends

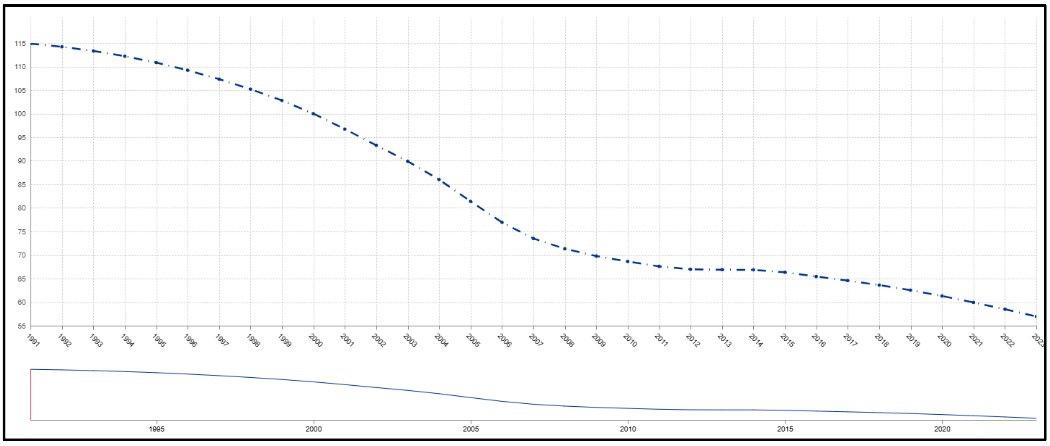
Current Status of European Pollinators

The IUCN European Red List of Bees (Nieto et al., 2014) shows that 9% of all bee species are threatened, with negative population trends for more than 45% of species, while 26% of bumblebees are classified as threatened. This primarily reflects the loss and fragmentation of semi-natural grasslands, and pressures from intensive agriculture and pesticides. Overall, 57% of all species were data deficient and could not be assessed at the EU level.

Since the 1950s, wild bees in the Netherlands, Belgium and Great Britain have generally declined in diversity and occurrence (Biesmeijer et al., 2006; Carvalheiro et al., 2013). There is evidence of parallel declines in bee species richness in many areas, along with declines in plant species that are reliant on insect pollinators in Great Britain or bee pollinators in the Netherlands, when compared to declines of wind- or water-pollinated plant species. Of the 137 UK wild bee species analysed between 1980 and 2016, 37% declined in occupancy and 20% increased, while the average trend across the species was a 25% decline. The declines were greatest between 2006 and 2013, and the average trend across species has since stabilised (Powney et al., 2019). Since 1909, 20 bee and wasp species have become extinct in Britain (Ollerton et al., 2014).

Although trends cannot currently be extrapolated for many pollinator groups or at regional scales (Potts et al., 2021), strong evidence shows that declines in bees and hoverflies reported in the European Red Lists and national studies are expected to continue across most EU agricultural land. Therefore, the EU Grassland Butterfly Index (https://data.europa.eu/data/datasets/) is commonly used as a proxy for overall pollinator decline in Europe.

Butterflies are extensively monitored across Europe through the European Butterfly Monitoring Schemes (https://butterfly-monitoring.net/). They occupy similar habitats and are highly sensitive to the same environmental pressures as many other pollinator groups, and their trends therefore mirror the responses of other pollinators for which comparable large-scale data are lacking (Van Swaay et al., 2022; 2025). The index integrates the population trends of 17 butterfly species monitored across the EU and is calculated from data from all 27 EU Member States, starting from 1990 until 2023. The European Grassland Butterfly Indicator 1990-2023 shows a linear decline of 32% in the EU27 and 36% in Europe in the period 2010-2020. The 2025 update of the European Butterfly Monitoring Schemes eBMS database (https://butterfly-monitoring.net/ebms-database) shows a 50.4% decline in the 1990-2023 period, equivalent to a -2.13% annual decrease (Van Swaay et al., 2025).



EU Grassland Butterfly Index 1990/2023 - EU aggregate from Eurostat data

The EU (2023) Impact Assessment Study to support the development of legally binding EU nature restoration targets assessed expected pollinator abundance trends if no additional action is taken. Extrapolations show that should the historical trend continue, the grassland butterfly indicator would likely decline by -23% by 2030, and by -48% by 2050, compared to 2017 values, meaning an average annual decline of -1.23%. This figure is consistent with the EU (2023) estimate of -1.5% annual degradation trend of agro-ecosystems (areas requiring restoration or conservation).

Reversing the decline of pollinators is an objective of the EU Biodiversity Strategy for 2030 (https://environment.ec.europa.eu/strategy/biodiversity-strategy-2030_en) and the EU Pollinators Initiative (https://environment.ec.europa.eu/topics/nature-and-biodiversity/pollinators_en).

The EU 2024 Regulation on Nature Restoration (https://green-forum.ec.europa.eu/nature-and-biodiversity/nature-restoration-regulation-implementation_en) paves the way for a broad range of ecosystems to be restored. The Grassland Butterfly Index is included among three indicators in agricultural ecosystems from which Member States must select at least two, with the obligation to put measures in place which aim to achieve an increasing trend in the chosen indicators by 2030 and thereafter until satisfactory levels are achieved.

Climate Change Impacts on Pollinators

Habitat degradation and climate change are globally acting as pivotal drivers of wildlife collapse, with mounting evidence that this erosion of biodiversity will accelerate in the following decades (Ghisbain et al., 2024; IPBES, 2019). Moreover, although climate change impacts to date are not likely to be among the greatest drivers, they are expected to increase (IPCC, 2020), together with indirect drivers such as increased irrigation and production of bioenergy. Projections of common farmland bird and grassland butterfly indicators suggest that these groups will continue to decline into the future, likely driven by agricultural practices and land use. Accordingly, conservation and restoration measures are likely to increase pollinator species diversity and abundance (Olmeda et al., 2019).

Climate change will play an important role in future trends, disrupting seasonal temperatures and precipitation patterns. Rising temperatures are expected to increasingly impact pollinators, influencing their geographic ranges, population sizes, and seasonal activity patterns (Hoegh-Guldberg et al., 2018; Rahimi et al., 2024; 2025), possibly leading to extinctions of some species at the southern and northern edges of ranges and in alpine species, whilst driving the expansion of other species to colonise new areas (EU, 2023).

Bumblebees are particularly vulnerable as cold-adapted species. The average temperature requirements of all bumblebee species do not reflect the broad climatic range of the EU but are instead concentrated in intermediate to cold conditions (Rasmont et al., 2015). For European bumblebees, this range extends from -1.6°C (Bombus hyperboreus) to 10.4°C (B. ruderatus), with a median of 7.0°C. This value, known as the Community Temperature Index (CTI) (Devictor et al., 2012), is commonly used to assess community-level changes in response to climate change.

Evidence from Species Distribution Models

The overall impact of climate change on many pollinator species is difficult to assess, as for most species we often lack the long-term data and mechanistic evidence needed to identify climate-driven declines and predict future trends (Kazenel et al., 2024).

The EU Status and Trends of European Pollinators (STEP) project (Potts et al., 2015) found that the vast majority of bumblebees (up to 46 species in 2050 and up to 52 species in 2100) will suffer from range contractions under high-end emission scenarios (3°C and 4.7°C). Results of two large-scale studies projected the potential future distribution in Europe for butterflies by 2080 (Settele et al., 2008) and bumblebees by 2100 (Rasmont et al., 2015). Under the extreme (+5.6°C) scenario, 16% of the modelled European bumblebees and 24% of the analysed butterflies are projected to be at extremely high climatic risk, 29% and 16% will be at very high risk, 46% and 30% are at high risk, 5% and 24% are at risk, and only 2% and 6% are at low risk. Only a limited number of species are projected to benefit from climate change under a full dispersal assumption.

Recent Multi-Species Projections:

Ghisbain et al. (2024) identified future large-scale range contractions and species extirpations under all future climate and land use change scenarios for 46 bumblebee species (Apidae, genus Bombus), for which long-term observation data, literature on biogeography and life history are available at the European spatial scale. Both environmental variables and occurrence data are used to generate ecological suitability through species-specific ecological niche modelling analyses that are then combined to compute an Ecological Suitability Index (ESI) and its evolution through time. Results show that temperature and precipitation have strong negative correlations with ecological suitability differences, suggesting that warming and altered rainfall patterns are major contributors to habitat loss, especially for cold-adapted species. Around 38-76% of studied European bumblebee species currently classified as 'Least Concern' are projected to undergo losses of at least 30% of ecologically suitable territory by 2061-2080 compared to 2000-2014.

Prestele et al. (2021) assessed the potential distribution of 47 European bumblebee species in 2050 and 2080 from existing European-scale distribution maps, based on a set of climate and land-use futures simulated through a regional integrated assessment model under three RCP-SSP scenarios. They found that direct climate impacts, although variable across species, dominate responses for most species, especially under high-end climate change scenarios (up to 99% range loss). Land use may amplify, attenuate, or offset changes to suitable habitat extent expected from climate impact depending on species and scenario, especially in low-end climate change scenarios (RCP2.6), indicating substantial potential for targeted conservation management.

However, many bee species are ecologically and evolutionarily different from bumblebees. Kazenel et al. (2024) documented 16 years of abundance patterns for a hyper-diverse bee assemblage in a warming and drying region, linking bee declines with experimentally determined heat and desiccation tolerances, and using climate sensitivity models to project bee communities into the future. Aridity strongly predicted bee abundance for 71% of 665 bee populations. Models forecasted declines for 46% of species and predicted more homogeneous communities dominated by drought-tolerant taxa, even while total bee abundance may remain unchanged. Such community reordering could reduce pollination services, because diverse bee assemblages typically maximize pollination for plant communities.

Rahimi et al. (2024; 2025) assessed the impact of climate change on pollinator habitat suitability and pollinator-dependent crops on a global scale due to climate-dependent variables. Their findings suggested that approximately 65% of bees are likely to witness a decrease in their distribution, with climate change's impact on bees projected to be more severe in Europe (average reduction of 56%) under SSP5-8.5. Climate change's anticipated effects on bee distributions could potentially disrupt existing pollinator-plant networks, posing ecological challenges that emphasize the importance of pollinator diversity, synchrony between plants and bees, and the necessity for focused conservation efforts. Under the most pessimistic climate change scenario (SSP5-8.5), they estimate that Europe is expected to see a 4.7% decrease in area suitability for pollination-dependent crops by 2070.

Pressures on Pollinators and Conservation Responses

The decline of both wild and managed pollinators is intrinsically linked to the broader degradation of agro-ecosystems, where pollination represents a key ecosystem service that is subject to the same underlying multiple pressures and drivers of impact (IPBES, 2016; Potts et al., 2015).

Key pressures include:

1. Land use change – conversion of natural and semi-natural habitats to intensive agriculture
2. Habitat loss and fragmentation – resulting from urbanization and infrastructure development, removal of hedgerows and field margins, loss of semi-natural grasslands, and decline of wild flowering plants
3. Environmental pollutants – especially systemic insecticides (neonicotinoids) and fungicides
4. Invasive alien species – though impacts remain difficult to generalize
5. Climate change – affecting phenology, distributions, and species interactions

These factors are most prevalent in intensive agricultural management and monoculture cropping systems. Habitat for pollinators has been drastically reduced due to agricultural intensification, leading to loss of natural and semi-natural habitats rich in wildflowers which pollinators use for foraging, nesting and hibernating (Biesmeijer et al., 2006; Potts et al., 2010; Goulson et al., 2015; IPBES, 2016; Zattara & Aizen, 2021). While it is difficult to quantify the loss of wildflowers from agricultural landscapes, herbicides have an indirect but significant impact on wild pollinators by removing the wildflowers on which they feed. The use of pesticides – linked to agricultural intensification – also has a direct and significant impact on pollinators (Nicholson et al., 2024).

Member State surveillance of the habitats in Annex I of the EU Habitats Directive shows significant losses of pollinator habitats in the period 2013 to 2018 (EEA, 2020; Kudrnovsky et al., 2020). Over half of the grassland habitats were assessed as in unfavourable-bad conservation status and nearly half had a declining trend over the period to 2018.

Conservation potential:

Remnants of natural and semi-natural habitats, hedgerows, and field margins, which supply essential flowering and nesting resources, can become important pollinator sources in different agro-ecosystems (IPBES, 2016; Winfree et al., 2009; Morandin & Kremen, 2013; Aviron et al., 2023). However, increasing distance from field edges into crop fields greatly reduces flower visitation and the number of visiting species. On average, bee visitation rates and richness are reduced by half at distances of approximately 670 and 1,500 m, respectively, from natural vegetation (Ricketts et al., 2008). As a consequence, not only does average crop yield often decrease with distance to field margins or natural vegetation (albeit at lower rates than pollinator abundance and richness), but it also becomes less predictable (Garibaldi et al., 2011).

This spatial relationship suggests that improving measures of conservation and restoration of semi-natural habitats within agro-ecosystems can reduce pollination loss. Expert estimates indicate that restoration of pollinator habitat on at least 10% of farmland will be the minimum needed to maintain the most common wild pollinator species (Dicks et al., 2015; Martin et al., 2019; Dicks et al., 2021; EU, 2023b).

Agro-Ecosystems status and trend

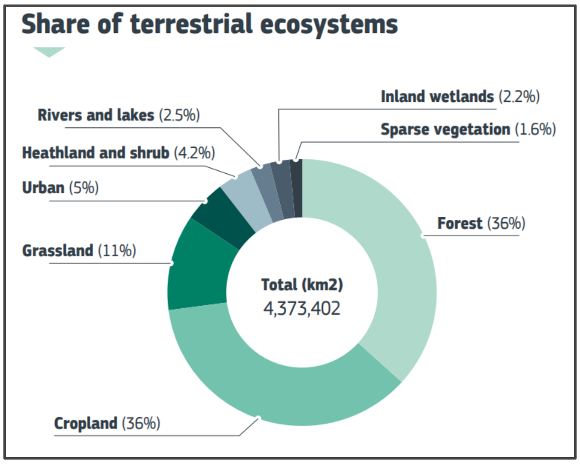
The capacity of ecosystems to provide services is closely linked to their extent and conservation status (Dasgupta, 2024). This is particularly evident for pollination, as pollinator species are intrinsically dependent on the condition and scale of their ecological niches, namely their specific habitats. Consequently, any evaluation of current or future ecosystem services, such as pollination, must be inseparable from an assessment of ecosystem definition, extent, and status.

The following section provides the necessary definitorial elements of the ecosystem framework analysis of our assessment, from an ecological and EU normative point of view.

The EU Habitats Directive lists 52 pollinator species in Europe, while Annex I includes habitat types significant for wild pollinators, such as the semi-natural grasslands (Kudrnovsky et al., 2020). Together with managed grassland and croplands, these habitats constitute the European Agro-Ecosystems and are of key importance for biodiversity as breeding sites for a variety of important pollinator species.

In 1992, the complementary European Habitats Directive (HD) was put in place, outlining similar protections for wild flora and fauna and their habitats. Special Areas of Conservation (SACs) were then drawn up and together, an ecological network of over 27 800 sites called Natura 2000 was formed. It comprises Sites of Community Importance (SCIs) and Special Conservation Areas (SAC) designated under the Habitats Directive and Special Protection Areas (SPAs) designated under the Birds Directive (referred to collectively as Natura 2000 sites). As an ongoing process, sites continue to be created and expanded upon. As a result of their high biodiversity value, and because many are now scarce and/or declining, most natural and semi-natural agricultural habitats in the EU are listed in 35 HD Annex I habitat types. Many associated species are listed in HD Annex II (197), IV or V or BD Annex I due to their need for special conservation measures. As of 2020, 81% of habitats and 63% of species covered by the HD are in an unfavourable (poor or bad) condition (EU 2023, b).

Agro-Ecosystems constitute a specific terrestrial ecosystem category, which encompasses many (HD) habitats and other habitats not included in the Habitat Directive. It includes all grasslands (temporary and permanent) and some other semi-natural habitats, which are usually grazed by livestock and/or used for other agricultural / silvicultural purposes, as well as all croplands including arable, vegetable, fruit and other permanent crops.



Share of habitat surface in EU terrestrial ecosystems from Natura2000 and Corine Land Use.
Source: JRC Publications Repository - EU Ecosystem Assessment

Agro-ecosystems can be generally divided into two broad categories, depending on the level of human management on them: 1) Natural and semi-natural agricultural habitats (many of which are HD Annex I habitats) and 2) Agriculturally improved grasslands and croplands (NOT Annex 1).

Permanent grassland and other habitats grazed by livestock					Crops			
Habitat types	Natural habitats	Semi-natural habitats		Modified grassland	Cultivated		Permanent	
		Pastures	Meadows	Conventional	Extensive	Intensive	Extensive	Intensive
HD Annex I habitats[234]	63							
BD Annex I birds[235]	54			32			5	
HD Annex II Butterflies[236]	9	25		0	0	0	0	0
European threatened amphibians[237]	3	5		0	1	0	0	0
European threatened reptiles[238]	1	4		0	0	0	4	0
Overall biodiversity importance	Very high, many species are restricted to such habitats	Very high, these habitats tend to be species-rich and declining; some species are restricted to such habitats and dependant on specific agricultural practices		Moderate, species diversity is much reduced compared to natural and semi-natural habitats, but some species of conservation importance use such habitats, sometimes in important numbers	High, such habitats are now rare and support some threatened species (esp. birds)	Low, especially in intensive farmland dominated landscapes, but biodiversity levels can be enhanced by appropriate measures	Moderate - High, such habitats are declining and support some threatened species	Low, especially in intensive farmland dominated landscapes, but biodiversity levels can be enhanced by appropriate measures

Agro-ecosystems in the EU and their importance for EU protected habitats and biodiversity. Source: EU (2023)(a) based on Poláková et al (2011)[2]

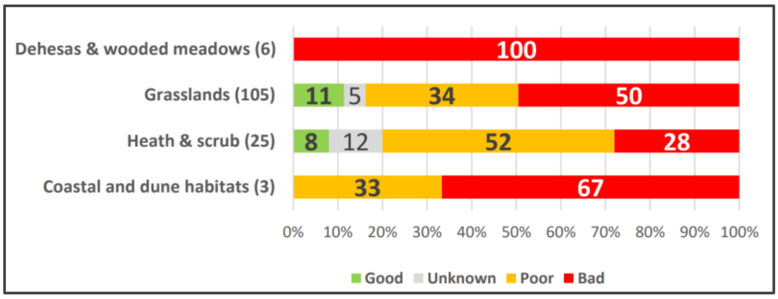
Detailed Agro-Ecosystems definition and mapping : MAPPING ECOSYSTEMS.xlsx

A) Natural and semi-natural agricultural habitats (HD Annex I habitats)

- Natural habitats: grasslands, shrublands and other habitats that can be extensively grazed but not significantly changed by livestock grazing or other human activities.
- Semi-natural agricultural habitats: vegetation and associated species that have not been planted and are dominated by native species, but are the result of human activities (woodland clearance, grazing and burning). Es: Grassland and shrubland pastures that are dependent on livestock grazing for their maintenance; Meadows that are dependent on mowing.

Article 17 reporting data for 2013-2018 for Annex I agricultural habitats (covering a part of natural grasslands and not covering modified grasslands and croplands). The 35 habitat types selected cover close to 177 442 km2 (4.5 % of the EU terrestrial area, except Romania). Out of the total area, over 43% is estimated to be inside the Natura 2000 network (about 77 025 km2). Agro-ecosystems represent on average 38% of the total surface area of the Natura 2000 network in the EU (EEA, 2010), and Natura 2000 sites contain 10.6% (or 22.2 million ha) of the total agricultural land of the EU-27.
EU 2023 (b) provides an assessment on their conservation status according to the 'best estimate' of degradation, which is based only on HD Annex I 2013-2018 assessments that reported on habitat conditions. The vast majority (84%) over 2013 - 2018 of the 35 HD Annex I (Natura 2000) agricultural habitats and grasslands at the EU level have an unfavourable conservation status (36% poor and 48% bad), with 45% having a deteriorating trend. Only 10% have a good conservation status. Concerning 2013/2018 trends, the group with the worst conservation status trends is 'grasslands' (52% deteriorating). However, 35% of the total area of the habitat area is reported as in 'unknown' (or not reported) condition. The European Environmental Agency (EEA) estimates that in 2018 at least 2 431 km2 would need to be re-created to achieve a 'favourable area', including for 2 145 km2 for grasslands, 183 km2 for temperate heath and scrub. However, the actual area that needs to be re-created is expected to be much higher since Member States only provided quantitative assessments of their 'favourable reference areas' (FRAs) for 48.1% of HD Annex I agricultural habitats, so that the estimated re-creation area increases to 3 743 km2.

2 Poláková, et al (2011) Addressing biodiversity and habitat preservation through Measures applied under the Common Agricultural Policy. Report prepared for DG Agriculture and Rural Development, Contract No. 30-CE-0388497/00-44, Institute for European Environmental Policy, London



Conservation status of Annex I agricultural habitats in the EU in 2018.
Source: EU (2023)(a) based on EEA data.

B) Agriculturally improved grasslands and croplands (NOT Annex 1)

This category comprises Agriculturally Improved Grasslands (which have been modified to increase agricultural productivity) and Cultivated Croplands (both intensively cultivated and organic/High Value Farming). These can be associated with the following CORINE Land Cover (CLC) classes:

- Regularly cultivated land, which includes non-irrigated arable land (CLC class 211), permanently irrigated land (212), rice-fields (213), vineyards (221), fruit trees and berry plantations (222), olive groves (223), pastures (231) and annual crops associated with permanent crops (241).
- Mixed cultivated land: complex cultivation patterns (242), agricultural area with significant areas of natural vegetation (243) and agro-forestry areas (244).
- Semi-natural areas with possible extensive agriculture practices: natural grasslands (321), moors and heathland (322), sclerophyllous vegetation (323).

Farmland also makes up around 40% of the total area included in Natura 2000 (High Nature Value farmland, HNV). Because a high level of biodiversity usually coincides with low agricultural productivity, most of the farmland in Natura 2000 is located in the more marginal farming areas (alpine meadows and pastures). In some of these areas, existing farming systems and practices are already likely to be compatible with the conservation of the species and habitats for which the site has been designated under Natura 2000 (ISBN 978-92-79-95905-9).

The precise area of natural and semi-natural agricultural habitats not covered by HD Annex I habitats is not known, as they have not been defined and mapped. According to Corine Land Cover data the total area of agro-ecosystems in the EU was 2 096 616 km2 in 2018 (48% of the EU terrestrial area). Whilst the Annex I (Natura 2000) data and Corine data are not directly comparable, approximately 1.9 million km2 are non-Annex I agricultural habitats.

	Area	2000	2006	2012	2018
Grassland	*km²*	504 778	503 411	502 454	500 566
	% share of total	11.50%	11.50%	11.50%	11.40%
Cropland	*km²*	1 604 793	1 600 732	1 596 331	1 596 050
	% share of total	36.60%	36.50%	36.40%	36.40%
Total agricultural area	*km²*	2 109 571	2 104 143	2 098 785	2 096 616
	% share of total	48.10%	48.00%	47.90%	47.80%
Total terrestrial ecosystems	*km²*	4 381 107	4 381 117	4 381 117	4 381 117
	% share of total	100%	100%	100%	100%

Areas from CORINE Land Cover for the years 2000-2018 as a percentage of the total EU terrestrial ecosystem area.
Source: EU (2023)(a) based on CLC data.

Despite specific CAP requirements, there is no formal requirement for reporting on their condition. However, numerous scientific publications have documented widespread and substantial declines in farmland biodiversity over at least the last 50 years (Voříšek et al. 2010; Tsiafouli, M. A. et al. 2014; Underwood at al., 2020; Reif et al. 2019).

Data sources:

- Natura 2000 (2022) : EEA geospatial data catalogue

Compilation of the data submitted by the Member States of the European Union. County and spatial level data on Annex I of Directive 92/43/EEC: habitat type, cover (ha) Conservation status, impact, pollution, species contained, management conservation measures.

- MAES 2018 area. Area under different ecosystem type https://www.eea.europa.eu/themes/biodiversity/mapping-europes-ecosystems
- CORINE Land cover and change accounts 2000-2018
- Maps and charts | European Environment Agency's home page
- EU 2023 (a)(b)

Agroecosystems and their services (in particular pollination) are threatened by increased pressures from climate change and agricultural practices, which affect overall ecosystem condition and processes as well as determining declining trends in biodiversity.

According to Natura 2000 (2022) (**EEA geospatial data catalogue**), four groups of pressures were reported affecting HD Annex I protected agricultural habitats (modified grasslands and croplands NOT included) :

• Habitat management with over 42% of all pressures; these include inadequate agricultural practices like intensive grazing (18%), under grazing (12%), forestry like logging and removal of dead and old trees (44%); however, over half of these pressures result from abandonment of grassland management (49%)

• Conversion and land use change amounts to 24%; from these, over one-third (35%) is originated from agriculture intensification, conversion to forest (23%), and construction of urban, industrial and leisure sites (28%)

• Natural processes, with over 15%; this is mainly due to natural succession (83%), which is often due to abandonment of agricultural activities

• Pollution, reported in 10% of assessments, mainly originating from agriculture (73%) or from mixed sources (25%).

Reporting under the Habitats Directive on pressures facing Annex 1 grasslands and agro-ecosystems only 1.5shows that only 1.5% of overall reported pressures are due to climate change.

Regarding NON Annex 1 modified grasslands and croplands habitats, evidence suggests that agricultural practices have the most significant impact on population trends in most farmland specialist species, especially those of improved grasslands and intensive crops:

• Fertiliser use: promotes dense, fast-growing grasslands and crops, but reduces plant diversity and associated animal communities.

• Ploughing and reseeding with grass cultivars: further reduces plant diversity, harms soil biodiversity, and favors uniform, high-density grasslands.

• Livestock Management and High grazing densities: decrease plant and structural diversity in pastures, lower animal abundance and diversity, and cause major losses of ground-nesting birds.

• Harvesting Practices like Silage cutting: leads to the loss of species-rich hay meadows and kills ground-nesting birds and other animals.

• Agrochemical Use. Herbicides/weed control: reduce plant species diversity in grasslands and crops, affecting dependent animal communities. Pesticides: diminish the abundance and diversity of invertebrates, which are key food sources for birds, bats, and other predators.

• Crop specialisation and limited rotations: reduce landscape heterogeneity, breeding and feeding options, and ecological connectivity.

• Changes in sowing timing: alter food availability in winter and affect vegetation structure needed for breeding.

• Habitat Loss and Simplification due to removal of boundary and non-farmed habitats (hedgerows, stone walls, terraces, ditches, woodlots, trees, ponds): decreases habitat diversity and reduces connectivity across landscapes.

Among most important generic impact drivers on the overall present and future status of Agro-Ecosystems the following trend can be identified:

1) Land Use Change

Based on trends over the last decade and foreseen trends in key land use defining indicators, no major changes are usually expected in the short term in EU agro-ecosystem extent in comparison to the current situation (EU agricultural outlook 2021-2031). However, estimates from the EU Joint Research Centre (JRC) and published in 2018 suggested in 2030 more than 11% of agricultural land in the EU would be at high or very high risk of abandonment (Agricultural land abandonment in the EU by 2030). The total area of land simulated to be abandoned over the period 2015-2030 is about 2 800 km2 per year on average (42 000 km2 by 2030). Arable land is projected to account for the largest share of this (70%), followed by pastures (20%) and permanent crops (7%). Almost 25% of the abandoned land area is projected to occur in mountainous areas, with arable again making up the majority (70%) of the abandoned mountainous land area. However, projected conversion dynamics of abandoned agricultural land suggests that only around 6 000 km2 would be transformed into forest or natural areas.

2) Climate Change impact on terrestrial biomes

According to IPCC, 2020 (SR15), constraining global warming to 1.5°C rather than 2°C has strong benefits for terrestrial and wetland ecosystems and their services (high confidence). These benefits include avoidance or reduction of changes such as biome transformations, species range losses, increased extinction risks (all high confidence) and changes in phenology (high confidence), together with projected increases in extreme weather events which are not yet factored into these analyses (Section 3.3). All of these changes contribute to disruption of ecosystem functioning and loss of cultural, provisioning and regulating services provided by these ecosystems to humans. Examples of such services include soil conservation (avoidance of desertification), flood control, water and air purification, pollination, nutrient cycling, sources of food, and recreation.

3) Pesticides

Over the period 2011 to 2019, sales of pesticides have oscillated increasing from around 360,000 t in 2011 to over 370,000 t in 2017, then falling to 333,000 t in 2019. The EU Ecosystem Assessment concluded that this trend to 2017 was stable (MAES, 2020) and a linear regression of the data spanning the period 2011 to 2019 suggests no significant trend.

Year	Pesticide sales (tonnes)	Utilised Agricultural Area (km^2)	Pesticide sales per UAA (t/km^2)
2011	358 873	1 793 654	0.200
2012	347 411	1 781 978	0.195
2013	351 184	1 780 988	0.197
2014	369 081	1 783 927	0.207
2015	366 630	1 789 956	0.205
2016	371 101	1 787 632	0.208
2017	359 913	1 788 222	0.201
2018	354 597	1 791 446	0.198
2019	333 418		

Total annual sales of pesticides in the EU-27 in the period 2011-2019. Source: EU (2023) based on Eurostat data.

There are specific targets in the Biodiversity Strategy and CAP targeting pesticide use, which can be expected to take effect by 2030 and further by 2050. Impacts from pesticide use are expected to decline as a result of the proposed measures to reduce the use of harmful pesticides, and to increase organic farming to 25% of the EUs agricultural land by 2030.

4) Fertilizers (nutrient runoff)

Agricultural land is the biggest source of nutrient pollution in water. While critical loads of nitrogen have been decreasing, the absolute values of all these pressures remain too high, and further reductions are needed. The assessment points to potential risks to ecosystems of continued nitrogen pollution in combination with other pressures, in particular climate change. For example, increased frequency and intensity of heavy precipitation events will facilitate run-off, increased temperatures will increase emissions to air from manure, and increased frequency and severity of droughts will increase nutrient concentrations in surface water bodies through reduced dilution.

At the EU level, gross nitrogen levels on agricultural land have been quite stable over the past decade at around 50 kg/ha/year while phosphorus balance has decreased by 0.25kg/ha/year from 2004 to 2015. The total consumption of nutrients in fertilisers increased between 2010 and 2017 by 11.6% for nitrogen and 14.8% for phosphorus. The trends in both phosphorus and nitrogen consumption, calculated using a linear regression against time, are statistically significant ($p < 0.05$) and adjusted-r 2 values of 0.6 for phosphorus consumption and 0.77 for nitrogen consumption.

	Consumption of P (t)	Consumption of N (t)	UAA (km2)	Consumption of P per UAA (t/km2)	Consumption of N per UAA (t/km2)
2010	1 060 420	10 401 860	1 801 368	0.589	5.774
2011	1 143 938	10 885 730	1 793 654	0.638	6.069
2012	1 097 467	10 384 633	1 781 978	0.616	5.828
2013	1 194 772	10 914 914	1 780 988	0.671	6.129
2014	1 184 716	11 141 635	1 783 927	0.664	6.246
2015	1 158 821	11 412 237	1 789 956	0.647	6.376
2016	1 183 657	11 278 535	1 787 632	0.662	6.309
2017	1 217 562	11 611 071	1 788 222	0.681	6.493

Total annual consumption of Phosphorus and Nitrogen in inorganic fertilisers in the EU-28.
Source: EU (2023) based on Eurostat data

Improvements are recorded in a diminishing atmospheric nitrogen deposition and nitrogen concentration in groundwater, but overall nutrient levels remain too high. Pressure on agroecosystems from deposition of pollution causing eutrophication or acidification, and from excess fertiliser application is likely to continue to decline. These sources of pollution have been reducing over the last two decades and planned policies and actions are likely to accelerate such declines thanks to specific targets in the Biodiversity Strategy and CAP targeting fertilisers and pesticides use, which can be expected to take effect by 2030 and further by 2050.

- Nutrient enrichment and eutrophication in Europe's seas | European Environment Agency's home page
- Nutrient pressures in European fresh and coastal water
- Risk of eutrophication measured as exceedance of critical loads of nitrogen deposition in Europe, in 2022 | European Environment Agency's home page
- Eutrophication caused by atmospheric nitrogen deposition in Europe | European Environment Agency's home page
- Joint Research Centre Data Catalogue - Nitrogen and phosphorus concentrations in European. - European Commission
- Joint Research Centre Data Catalogue - Nitrogen and phosphorus loads to the sea (1990-201... - European Commission
- European Nitrogen Assessment (ENA) | Nitrogen in Europe

EU (2023) provides future projections of the overall Agro-Ecosystem area requiring management, restoration and re-creation respect to 2018, taking into account the expected rates of habitat loss. It is very likely that without interventions the condition of agro-ecosystems would continue to decline in the next three decades.

If no further actions are taken, the best estimates assess that in 2030 the 27% of agro-ecosystems area will require management and restoration practice due habitat degradation, while the ecosystem loss (area requiring complete restoration) is 1.5% per year. This loss trend is derived using as a proxy the High Nature Value Farmland area in the EU from CLC accounting layers, assuming degradation levels to 2030 for HD Annex I agricultural habitats will not change. Main pressures are likely to continue, although the impacts from pesticide use are expected to decline as a result of the proposed measures to reduce the use of harmful pesticides, and to increase organic farming to 25% of the EUs agricultural land by 2030. Whilst climate change impacts to date are not likely to be among the greatest on Annex I habitats, they will increase, especially indirect impacts, such as from increased irrigation and production of bioenergy crops.

Projections of common farmland bird and grassland butterfly indicators suggest that these groups will continue to decline into the future, in part likely driven by changes in agricultural land use and cover. The land conversion and habitat modification discussed above are the major current pressures on Annex 1 habitats and it seems likely that this will remain the case in the coming decades.

Services provided by agro-ecosystems in good ecological condition	
Carbon sequestration	grasslands are net carbon sink
Wildfire control	grasslands, wet areas and buffer strips can be fire breaks
Water filtration / water retention	from nitrogen removal, increased infiltration rate
Natural pest control	presence of natural predators
Soil productivity / crop provision	increased soil fertility in HNV/organic farming
Flood control / soil erosion	reduced surface run off
Climate change impacts resilience	increased droughts resilience, wildfires and floods control
Pollination	semi-natural habitats and grassland are breeding sites

Increased ecosystem services provisioning with respect to intensively managed agricultural land and monoculture.
Source: our elaboration based on multiple sources.

Appendix 3. Caveats, novelty and differences with existing literature

A similar approach and goals of our framework is used by Johnson et al., 2023 GTAP-InVEST (GTAP-AEZ + InVEST + SEALS), which imposes a stylized partial collapse of ecosystem services, including wild pollination, to 2030, using the standard dependency ratios and a habitat-proximity pollination proxy, resolved economically at country/region level.

Another approach is the one of von Jeetze et al, 2023, which couples MAgPIE-SEALS to project pollination sufficiency (semi-natural habitat within a 2 km foraging buffer) at to 2050, with climate explicitly excluded from pollinator dynamics and no CGE at all.

Both approaches model pollination sufficiency as a function of habitat suitability and proximity to pollinator-dependent crops, operating at very fine spatial resolution. Yield losses are calculated at granular level, aggregated up to agro-ecological zones, and passed into GTAP, where they drive shifts in production, trade, and welfare. This framework excels at linking land-use change with economic outcomes but does not directly capture biodiversity dynamics under climate impacts.

Our approach, by contrast, relies on pollination potential projected changes using long-term biodiversity indicators and climate-driven species distribution models. It explicitly incorporates climate change scenarios. Several caveats follow from this design, most of which are shared with the literature on which we build. As in Johnson et al. (2023) and the broader pollination-economics tradition, we rely on crop-specific dependency ratios and represent pollination supply through a spatial proxy rather than a mechanistic measure of pollinator abundance or visitation; managed honeybees and active pollination management are not modelled, so our yield shocks are best read as upper bounds where such substitution is feasible. Exogenous land is required to preserve consistency between the SSP land-use realization that conditions the SDM pollination projections and the land allocation in the CGE; capturing reallocation in response to the pollination shock would require endogenous land–habitat–economy coupling, i.e., the feedback we deliberately exclude. Because shock-induced adaptation is omitted, and only the EU is shocked while the rest of the world is held constant, which may attenuate estimated losses through trade, the regional tail losses are upper bounds. Our contribution lies not in any single module but in their combination: to our knowledge, this is the first assessment to propagate a multi-model, climate-driven species-distribution ensemble through a sub-national (NUTS-2) general equilibrium model over a long horizon, complementing the stylized, land-use-driven and largely static representations of pollination in existing integrated assessments.